Semantic Chunking Techniques for Natural Language Processing

Jaya Sai Krishna Paleru
Department of Data Science
University of North Texas
Denton, TX, USA
JayasaikrishnaPaleru@my.unt.edu

Tarun Podila
Department of Data Science
University of North Texas
Denton, TX, USA
TarunPodila@my.unt.edu

Naveen Chowdary Vasireddy
Department of Data Science
University of North Texas
Denton, TX, USA
NaveenChowdaryVasireddy@my.unt.edu

Vishala Vadla
Department of Data Science
University of North Texas
Denton, TX, USA
VishalaVadla@my.unt.edu

***Abstract*—As digital text keeps growing rapidly, there's a greater need for NLP methods that can handle long documents. However, transformer-based language models struggle with long texts because of their limited context windows and high computational costs. To address this, breaking documents into smaller "chunks" has become an important step, especially in Retrieval-Augmented Generation (RAG) systems. Poor chunking can lead to more factual mistakes, hallucinations, and loss of meaning.**

This review looks at 26 recent studies (2018–2025) and finds that chunking methods have evolved from simple fixed-size or rule-based approaches to more advanced techniques. These include using semantic similarity, clustering, dynamic programming, LLM-driven chunking, adaptive chunk sizes, and hybrid methods. Despite these improvements, challenges remain—such as the lack of standard ways to evaluate chunking, inconsistent benchmarks, high computational demands, and varying results depending on the task or document type.

The review highlights the need for chunking strategies tailored to specific domains, better evaluation metrics, and collaboration across fields. Balancing efficiency, context, and meaning is key to improving long-document processing and RAG systems.

I. Introduction

The vast and continuous growth of digital information has increased the demand for robust Natural Language Processing (NLP) tools capable of effectively understanding and extracting meaning from large volumes of textual data. Recent advances, particularly Transformer architectures and Large Language Models (LLMs), have been highly successful across a variety of NLP tasks [1] [2]. Yet, a fundamental difficulty remains: these powerful models face considerable computational challenges when processing really long documents in one go. Specifically, Transformers are known to scale quadratically in computational requirements due to their self-attention mechanism, and generally utilize fixed context windows, restricting their direct ability to handle very long inputs [3] [4] [5] . Such constraints often had led researchers and practitioners to either truncate the input texts—risking significant information loss—or adopt more complex, resource-intensive sparse attention mechanisms designed to alleviate these constraints [4] [6].

To address these limitations, a preprocessing step known as "document chunking" or "segmentation," in which long texts are divided into smaller, more manageable pieces, has become really important [3] [7]. Chunking involves breaking down lengthy documents into segments manageable enough to be processed individually or sequentially by NLP models. This practice is particularly crucial within Retrieval-Augmented Generation (RAG) systems [8] [9]. RAG frameworks aim to improve the accuracy and factual grounding of LLM-generated content by incorporating external information—retrieval systems first fetch relevant document chunks based on a user query, and these selected chunks then inform and contextualize the generative process [8] [10]. Therefore, the initial chunking strategy has a significant impact on overall systems performance; poorly chosen segments/chunks could disrupt semantic meanings, reduce contextual coherence, and at last can also lead to selecting incomplete or irrelevant information. As a result, the quality and reliability of LLM outputs could degrade,possibly making factual mistakes worse and "hallucinations," where models give answers that sound believable but are actually wrong [9] [11] [12] [13].

Historically, chunking often uses simple heuristic rules, such as slicing text into fixed-size blocks using character or token counts, or dividing texts along obvious structural units like sentences or paragraphs [7] [14]. Although these basic approaches are computationally inexpensive, they frequently prove inadequate; arbitrary divisions can harm textual coherence, interrupt important context across boundaries, and disregard valuable structural information, especially for complex document formats like detailed reports or multi-column layouts [11] [12] [15] [16].

In response, the idea of "semantic chunking" has gained a lot of attention from researchers [7] [12] [17]. Instead of splitting text using fixed rules, this method divides text based on its actual meaning, producing chunks that make sense on

their own and keep their internal meaning and context [7] [17]. Several semantic chunking methods have been introduced, leveraging a lot of techniques such as semantic similarity measures between embeddings [7] [12], clustering approaches [17] [7] [18], interventions by language models themselves to detect semantic shifts [17] [18] [19], unsupervised keyphrase extraction methods [4], and hybrid approaches that combine semantic analysis with visual, spatial, or structural characteristics of documents [11] [16]. More recently, sophisticated methods also include embedding techniques specifically designed for maintaining context across contrasting chunks [20], as well as strategies that dynamically adjust the detail of chunking based on particular queries [21].

Despite the rapid growth and complexity of these various proposed techniques, current research lacks a proper overview or comparative analysis. There is still a big gap in bringing these approaches together, comparing their pros and cons, and giving clear, practical advice on how to choose the best chunking methods. Importantly, choosing the right chunking strategy depends a lot on factors like the type of document, the specific domain (e.g., legal [22] or financial [11]), the intended NLP task or use-case (such as summarization [5], question answering [15], or instructional design [23]), and requirements for computational efficiency [24]. Also, the best chunking strategy usually depends a lot on the specific task. Deciding between better performance and the amount of computational resources or complexity needed is still an important question [24] [25].

Therefore, this literature review aims to close this gap by carefully reviewing ,analyzing the current semantic chunking methods and By bringing together what is currently known about semantic chunking in research, this review aims to make the range of available methods clearer, helping researchers and practitioners choose and develop the best strategies for working with long documents.

II. Method

A systematic review of recent studies on semantic chunking methods in NLP is carried out in this work. The primary objective is to provide a concise and comparative overview of the existing techniques, addressing the lack of resources that examine, arrange, and assess these strategies for various use cases and document kinds. In order to ensure that the review is comprehensive and objective, a methodical approach was taken, adhering to the PRISMA guidelines [26] wherever applicable.

III. Research Questions

This systematic review is motivated by the need to consolidate current understanding and to provide comparative insights into the diverse range of semantic chunking techniques. Specifically, the study aims to address the following research questions, which have been formulated directly based on the gaps identified in the existing literature:

1) RQ1: What are the primary categories and characteristics of existing semantic chunking methods described in the current literature?
2) RQ2: What are the reported strengths and weaknesses of different semantic chunking approaches?
3) RQ3: How effective are specific semantic chunking methods reported to be for different document types (e.g., narrative, legal, financial, technical) and downstream NLP tasks (e.g., RAG, summarization, classification, information extraction)?
4) RQ4: What influence do factors such as document domain, structural complexity, and specific task objectives have on the performance and suitability of various semantic chunking techniques?
5) RQ5: What are the key evaluation methodologies and metrics used to assess the performance of semantic chunking techniques?

IV. Inclusion and Exclusion Criteria

To ensure the focus and relevance of this systematic review, specific inclusion and exclusion criteria were established prior to the literature search. The inclusion criteria were as follows:

- **Language:** English- language publications.
- **Publication type:** Peer-reviewed journal articles, conference papers, reputable preprints (e.g., arXiv), relevant dissertations, and technical reports.
- **Content focus:** Primarily addressing semantic chunking techniques within NLP or information systems, with substantial coverage of methodologies, implementation details, applications (e.g., Retrieval-Augmented Generation, summarization, information retrieval), comparisons, evaluation frameworks, or factors influencing effectiveness.
- **Time frame:** Publications from January 2018 to December 2025, to capture recent developments.

Conversely, the exclusion criteria were defined as:

- **Language:** Non-English studies.
- **Publication type:** Editorials, commentaries, opinion pieces, abstracts lacking full texts, patents, and non-technical blogs or news articles.
- **Content focus:** Studies that minimally mention chunking or segmentation, or address unrelated NLP topics (such as image or speech segmentation).
- **Relevance:** Studies limited exclusively to non-semantic chunking techniques (e.g., exclusively fixed-size or simple rule-based methods) without comparison or context related to semantic approaches.
- **Time frame:** Publications outside the specified range (pre-2018 or post-2025).

V. Databases and Search Method

A comprehensive search was conducted across multiple academic databases to locate pertinent literature. The primary databases that were searched were Web of Science, arXiv, Scopus, and the ACL Anthology. Google Scholar was used as

an extra resource to locate any publications that might have gone unnoticed.

The search strategy employed a range of keywords related to the primary concepts. ”Semantic chunking,” ”text segmentation,” ”document chunking,” ”natural language processing,” ”large language models,” ”retrieval augmented generation,” ”transformer models,” ”long document processing,” ”evaluation metrics,” ”chunking strategies,” and ”text partitioning” were some of the keywords. These terms were effectively combined using boolean operators (AND, OR) within the search interface of each database. For example, a standard search string might look like this: (”semantic chunking” OR ”semantic segmentation” OR ”document chunking”). AND (”natural language processing” OR ”LLM” OR ”RAG”). Searches were limited to publications within the specified time frame (2018-2024) and restricted to the English language. The search string we have used and more information about the strategy can be found in supplementary file 1 [27].

VI. Study Selection

The study selection process followed a multi-stage screening approach. Initially, the database searches yielded approximately 490 records. After removing the duplicates the count came down to 370.

- **Title and abstract selection:** Two reviewers independently screened the titles and abstracts of these records against the inclusion and exclusion criteria. Records clearly not meeting the criteria (e.g., wrong topic, wrong publication type) were excluded [28].
- **Full- text review:** The full texts of these 270 articles were retrieved and assessed independently by the two reviewers for eligibility based on the full inclusion and exclusion criteria. Any disagreements about inclusion were resolved by discussion or by consulting a third reviewer when necessary.
- **Final selection:** After the full-text review, approximately 50 studies were deemed eligible for quality assessment.

VII. Quality Assessment

The methodological quality of the 50 studies deemed eligible after the full-text review was assessed based on following criteria:

- **Clarity of research aims:** Were the goals of the study clearly stated?
- **Appropriateness of methodology:** Was the described chunking method clearly explained and appropriate for the stated aims?
- **Evaluation rigor:** If evaluated, was the evaluation methodology sound (e.g., appropriate datasets, metrics, baselines)?
- **Clarity of results:** Were the findings presented clearly and related back to the research aims?
- **Contribution to the field:** Did the study offer a meaningful contribution to the understanding or application of semantic chunking?

Studies were categorized qualitatively as ’High’, ’Moderate’, or ’Low’ quality. Those rated as ’Low’ quality (for example, studies showing significant methodological flaws, unclear descriptions, or unsubstantiated claims) were excluded. Ultimately, 26 studies with ’Moderate’ or ’High’ quality ratings were included in the synthesis phase of this review.

VIII. Data Extraction and Synthesis

For the 26 studies included in the final review, data were systematically collected using a structured extraction form developed to address the research questions. Two reviewers independently performed data extraction, resolving disagreements through discussion and consensus.

The data extraction form captured the following key details:

- Author(s) and publication year.
- Stated objectives or problem addressed.
- Type of semantic chunking method proposed or discussed (e.g., embedding-based, model-based, rule-based, hybrid).
- Core principles and algorithmic details of the chunking method.
- Implementation specifics (e.g., models used, parameters)
- Application domain(s) and specific downstream NLP task(s) targeted (e.g., RAG, summarization, classification).
- Document types considered (e.g., general text, legal, financial, scientific).
- Evaluation methodology, datasets, and metrics used.
- Reported performance, results, and key findings.
- Identified strengths and advantages of the method.
- Identified weaknesses, limitations, or challenges.
- Comparisons made with other chunking techniques.

After data extraction, a thematic synthesis was conducted. The collected data were organized according to the research questions, identifying common themes, patterns, contrasting results, and trends among the studies. Specifically, the synthesis examined types of chunking methods, their effectiveness across various contexts, what are factors influencing their performance, evaluation practices, and reported strengths and weaknesses for all the strategies. This analysis aimed to build a clear, integrated narrative addressing each research question based on the evidence gathered from the included studies.

IX. Results

During the initial search, 490 documents were found across databases, blogs and technical reports.After eliminating duplicates, 370 were left in the identification phase.Following screening based on title and abstract 270 articles were retrieved.Finally, 26 studies met the eligibility criteria, as outlined in Figure 1 depicting the study selection process. [htbp]

X. Articles Quality and Assessment Results

Based on the qualitative assessment using the five specified criteria (Clarity of Research Aims, Appropriateness of Methodology, Evaluation Rigor, Clarity of Results, Contribution to the Field),Seventeen studies (approximately 65.38%)

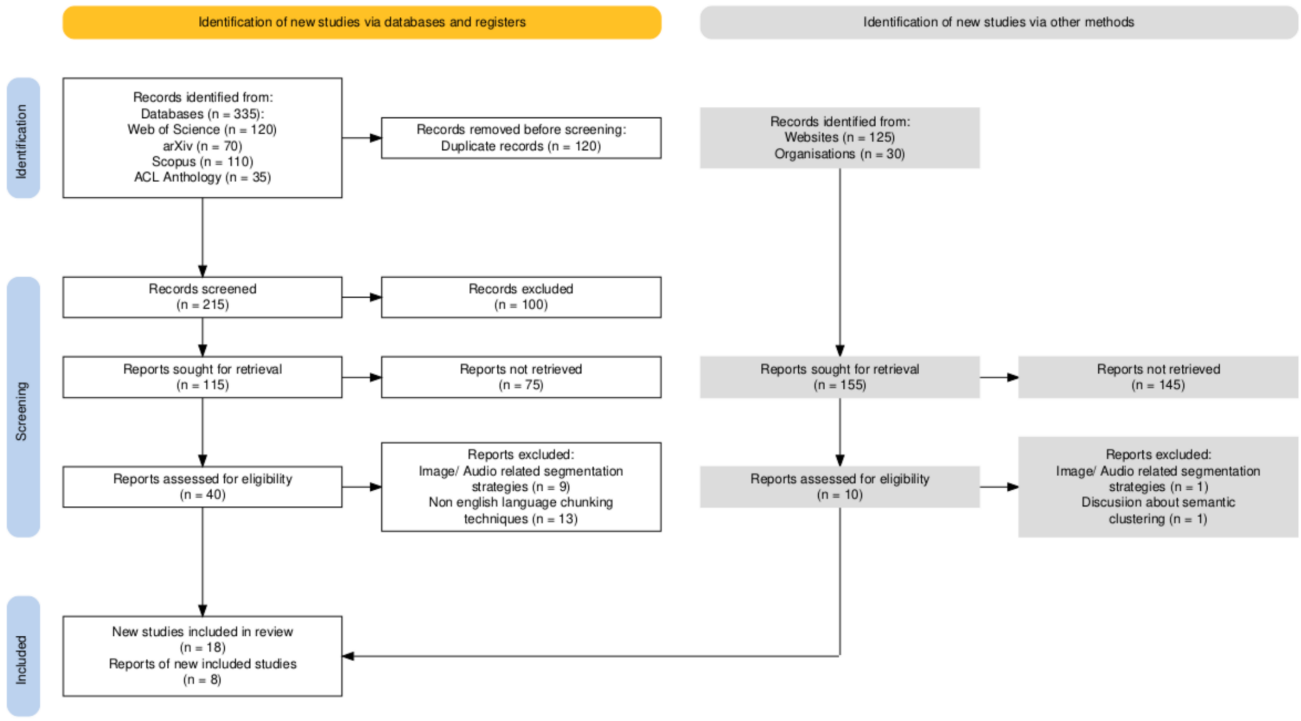

Fig. 1. Article selection.

maintained a 'High' quality rating, demonstrating strong alignment with all criteria including methodological rigor and contribution. Four studies (approximately 15.38%) were assessed as 'Moderate' quality, typically representing older foundational works or studies with less rigorous evaluation relative to recent advancements. Five studies (approximately 19.23%), primarily blog posts or web articles, were categorized as 'Low' quality under the stricter criteria, mainly due to lacking formal methodology, rigorous evaluation, or novel contribution, although they may still offer valuable conceptual explanations.For further details on the qualitative assessment process and individual study ratings, refer to the supplementary file detailing the quality assessment [28].

XI. Thematic Review of Document Chunking Techniques

A. The Evolving Landscape of Document Chunking in an Era of Large Language Models

The digital age has brought a huge amount of text-based information, making it important to have Natural Language Processing (NLP) systems that can handle, understand, and make use of large collections of documents. Transformer architectures [3] and the Large Language Models (LLMs) built upon them [9] have become leaders in this area. These models have shown impressive results in many NLP tasks. However, using them on very long documents is often challenging due to basic limits in computing power and model design. Transformers rely on self-attention mechanisms, which usually require computing resources that grow quickly as the length of the input text increases. Additionally, most models have a fixed limit on the length of text they can process at once, making it difficult to handle very long documents in one go [5] [12] [17]. Because of this limitation, users often face a tough choice: either shorten the input text and risk losing important information, or use more complicated and resource-heavy methods like sparse attention [16], which may not always be easy to implement or suitable for every situation.

To deal with these issues, document chunking—also known as text segmentation—has become an essential step before processing texts [5] [29]. This involves carefully splitting long documents into smaller, easier-to-handle pieces called "chunks," which NLP models can then process separately or one after another. Effective chunking is especially important in Retrieval-Augmented Generation (RAG) systems [30] [25]. RAG systems aim to improve the accuracy and factual correctness of content generated by large language models by pulling in relevant external information. They do this by first using a retriever component to find and select useful chunks from a collection of documents based on a user's query. These chunks then help guide and provide context for the language model as it generates responses [18] [25]. Because of this, the way documents are initially chunked can greatly affect how well the entire RAG system performs [11] [31] [18] [20].

In fact, poor chunking can cause a chain reaction of problems, greatly lowering the quality of the information retrieved and, as a result, the quality of the generated output. Random or badly planned splits can break the flow of meaning, separating related ideas or evidence into different chunks [31] [20] [21]

[22]. This fragmentation can confuse the retriever, causing it to pick incomplete or less meaningful chunks. As a result, the language model may struggle to produce accurate, relevant, and clear responses. This can even make existing weaknesses of language models worse, such as factual mistakes or "hallucinations"—statements that sound believable but are actually incorrect [15] [32] [30] [21]. Because of these issues, researchers have gradually moved away from simple, rule-based chunking methods toward more advanced approaches that consider meaning and context. These newer semantic and dynamic chunking methods aim to better maintain the original meaning and coherence of the text, and they form the main focus of this review.

B. Foundational Chunking Strategies: Simplicity and Its Trade-offs

The earliest methods for document chunking focused mainly on being simple and easy to compute. These basic approaches are still sometimes used today, but they mostly serve as benchmarks to compare against newer, more advanced techniques.

Fixed- sized chunking is probably the simplest method, where the text is divided into chunks of a set length, usually measured by characters or tokens [14] [29]. To avoid cutting off information too abruptly at chunk boundaries, chunks often overlap slightly with each other [3] [18]. The main benefit of fixed-size chunking is that it's easy to implement and doesn't require much computing power [24]. However, a major downside is that it doesn't consider the meaning or structure of the text. Random splits can cut sentences in half, separate related ideas, or break apart sections that should stay together, resulting in chunks that don't make sense or lose important context [3] [12] [18]. The effectiveness of fixed-size chunking depends heavily on choosing the right chunk length and overlap size, which usually requires trial-and-error adjustments for each specific dataset or task. Because of this, the chosen settings might not work well across different types of documents [24].

Going beyond simple fixed-size splits, **rule-based chunking (syntactic/ structural** uses clear textual markers or structural hints to divide documents into chunks. Common examples include splitting text at sentence endings (like periods, question marks, or exclamation points) or at paragraph breaks [7] [29]. More advanced rule-based methods might use patterns, such as headings or bullet points, as natural places to split the text [22]. The main advantage of these methods is that they take advantage of the structure authors use to organize their writing, often resulting in chunks that feel more natural and coherent compared to fixed-size methods [29]. However, relying on punctuation or formatting can be problematic, especially if the text is messy or formatted inconsistently. More importantly, these methods often miss meaningful boundaries that aren't clearly marked by punctuation or formatting. As a result, chunks may be grammatically complete but still lack semantic coherence or become too large [7] [30]. Additionally, creating effective and widely applicable rules can require significant effort and may need to be tailored specifically to each domain.

Because these basic methods have clear limitations—especially their difficulty in consistently creating chunks that preserve meaning—researchers began developing more advanced chunking approaches that pay closer attention to the actual content and meaning of the text.

C. Semantic Chunking: Striving for Meaningful Segmentation

To overcome the weaknesses of simple structural or fixed-length chunking, researchers developed semantic chunking methods. These methods aim to split documents based on their meaning and topic coherence, rather than just their structure or length [3] [29] [24]. The main idea is to create chunks that each focus on a single concept or topic, keeping the meaning clear and consistent within each chunk. This helps preserve context better and provides more useful input for NLP tasks such as RAG and summarization [5] [17] [14] [17] [18]. Semantic chunking includes several different approaches:

- **Embedding-based similarity:** A common method is to represent sentences, paragraphs, or other pieces of text as embedding vectors using pre-trained language models (such as Sentence-BERT or Universal Sentence Encoder) [14]. Then, the semantic similarity—usually measured by cosine similarity—between neighboring text segments is calculated. When this similarity falls below a certain threshold, it suggests a change in topic or meaning, and a chunk boundary is placed there [3] [7] [24]. This approach takes advantage of modern embeddings, which are good at capturing subtle meanings and relationships within text.
  - **Analysis:** This method is intuitive and directly focuses on changes in meaning. It can adjust to different types of content and easily detect shifts in topics that simpler, rule-based methods might overlook.
  - **Critique/ limitations:** The effectiveness of this method strongly depends on choosing the right embedding model and similarity threshold, which often requires careful tuning and may not work equally well across different document types or topics [3]. Computing embeddings and similarities can also be resource-intensive for very long texts [12] [22]. Additionally, a sudden drop in similarity doesn't always match where a human would naturally split the text, and gradual topic changes can be hard to detect.
- **Clustering approaches:** Another method uses clustering algorithms (such as K-Means or agglomerative clustering) on sentence embeddings or short text segments [7] [23]. Sentences with similar meanings are grouped together into clusters. Chunks are then created either by grouping consecutive sentences from the same cluster or by treating each cluster as a separate topic-based segment.
  - **Analysis:** Clustering can group sentences that share similar meanings, even if they're not next to each

other, helping to identify broader themes. This makes it useful for dividing texts into meaningful topic segments.
  - **Critique/limitations:** Choosing the right number of clusters (k) can be tricky and often needs to be decided beforehand. Also, the chunks created might not always be continuous in the original text, meaning extra steps are needed afterward to form coherent segments. Similar to embedding-based methods, clustering can also be computationally expensive, especially with large amounts of text.
- **Advanced semantic techniques:** More advanced methods have also been developed. For example, Semantic Self-Segmentation (S3) uses sentence embeddings combined with dynamic programming to identify the best places to split text, especially for tasks like abstractive summarization [5] [17] [33]. Other approaches involve training simpler sequence-tagging models (such as BiLSTM-CRF) to directly predict meaningful chunk boundaries by labeling segments as beginning, inside, or outside. These models can learn from labeled examples or use self-supervised signals [34].
  - **Analysis:** These methods can capture more complicated patterns of meaning and identify chunk boundaries more accurately than simpler methods like threshold-based similarity or clustering.
  - **Critique/ limitations:** Training these specialized models usually requires labeled data, which can be hard to find for chunking tasks, or advanced self-supervised methods. Also, dynamic programming techniques can still be computationally demanding.

D. Advanced and Dynamic Chunking Strategies: Pushing the Boundaries

Realizing the shortcomings of both traditional and typical semantic chunking methods, researchers have started developing more advanced and flexible approaches. These newer methods often make use of large language models (LLMs) themselves or include smarter ways to maintain context and adaptively split texts into meaningful segments.

- **Context- preserving techniques:** One major challenge in chunking is keeping context intact across chunk boundaries. Some advanced methods specifically tackle this issue. For example, Recurrent Chunking Mechanisms (RCM) [15] use reinforcement learning and recurrent connections to dynamically handle segments, allowing context to flow between chunks. This is especially helpful for tasks like Machine Reading Comprehension, where connections between segments matter a lot. Another creative method is "Late Chunking" [20], which first creates embeddings for the whole document using long-context models (like Longformer [6]) and then forms chunks based on these embeddings. This ensures each chunk is created with awareness of the entire document's context.
  - **Analysis:** These methods are a big improvement in reducing context loss, which is a major weakness of simpler chunking approaches. They allow the model to better understand the whole document before splitting it into chunks or help context move smoothly across chunk boundaries.
  - **Critique/ limitations:** The main downside is that these methods usually require more computing power and sometimes depend on specific, often larger, models designed for handling long contexts. Additionally, the performance of RCM can vary depending on how carefully the reinforcement learning process is set up.
- **LLM- Driven and adaptive granularity methods:** With the rise of powerful large language models (LLMs), researchers have started using these models directly to help with chunking. For example, LumberChunker [31] prompts an LLM to find natural points in paragraphs where the topic or narrative shifts, creating chunks of varying lengths that make sense on their own—especially useful for narrative texts. Another method, called Meta-Chunking or Perplexity-based (PPL) Chunking [32], uses an LLM's perplexity scores (which measure how well the model predicts upcoming words) to detect shifts in context, helping to identify good places to split the text. This can be more sensitive to subtle meaning changes than simpler embedding-based methods. Additionally, recognizing that the best chunk size can differ even within the same document depending on the task or query, Mix-of-Granularity (MoG) [21] uses a router network to dynamically choose the most suitable chunk size—such as sentences, paragraphs, or custom semantic segments—to improve performance in retrieval tasks involving diverse information sources. Some studies have also explored using LLMs to summarize sections or generate questions to help define chunk boundaries [35].
  - **Analysis:** These LLM-driven and adaptive methods are very flexible and can produce highly meaningful, context-sensitive chunks. They can pick up on subtle shifts in meaning or topic that simpler methods might overlook.
  - **Critique/ limitations:** The biggest downside is that these methods can be computationally expensive and slow, especially when repeatedly running LLM inference or processing very long documents. Their effectiveness can also depend heavily on how prompts are designed and the quality of the LLM being used. Additionally, training specialized parts like the router in MoG adds extra complexity.
- **Hybrid approaches:** To combine the advantages of different methods, hybrid approaches use both structural (or layout-based) information and semantic analysis. For example, S2-Chunking [16] is made for complex documents with rich visual layouts, such as scientific papers. It uses visual clues like columns, figures, and headings

along with semantic meaning to decide how to split the text. Similarly, in specialized fields like finance, chunking based on specific structural parts (such as tables or sections labeled "Risk Factors" or "Financial Statements") has been shown to work better than general paragraph-based or purely semantic methods, as demonstrated by Jimeno Yepes et al. [11]. These hybrid methods often first analyze the document structure and then apply semantic criteria within each identified structural section.

  - **Analysis:** Hybrid methods work especially well for structured or semi-structured documents, where layout and formatting give clear hints about meaning. By combining different types of information, these methods can produce more reliable and meaningful chunks.
  - **Critique/ limitations:** The main difficulty with hybrid methods is that combining different types of information (like text, visuals, and structure) can be complicated. These methods are often designed for specific domains or document types, meaning they usually need significant adjustments to work well with new kinds of documents or layouts. This limits how easily they can be applied to different situations without extra effort.

E. Domain- Specific Chunking Considerations: Tailoring to Context

Because textual data can vary greatly across different fields and document types, chunking strategies often need to be customized. A method that works well in one situation might not be effective in another. Each type of document has its own unique structure and meaning, which strongly affects how it should be split into chunks to get the best results in later processing steps.

- **Legal texts:** Legal documents typically have complex sentences, specialized vocabulary, many cross-references, and clear hierarchical structures (such as articles, sections, and clauses). General chunking methods often have trouble maintaining these important connections. Studies suggest that no single chunking method consistently works best for legal texts. Instead, hybrid methods that combine semantic meaning with the explicit structure of legal documents (like section headings or defined terms) tend to be more effective [22]. Usually, the main goal is to ensure that related legal provisions or arguments stay together within the same chunk.
- **Financial reports:** Financial documents, such as annual reports or SEC filings, usually have a clear structure. They often include narrative text, numerical data presented in tables, and distinct sections like "Management's Discussion and Analysis," "Risk Factors," or "Financial Statements." Research has found that chunking methods that use these structural elements tend to perform better than general approaches like paragraph-based or fixed-size chunking, especially for tasks like financial question-answering [11] [20]. For example, keeping entire tables or clearly marked sections as single chunks helps maintain the completeness and accuracy of the information.
- **Narrative texts:** For stories, novels, or news articles, good chunking usually matches the natural flow of the plot, scene changes, or shifts in themes. Simple methods like splitting by sentences or paragraphs can interrupt the narrative awkwardly. Methods that use large language models, such as LumberChunker [31], are especially promising here, as they can detect natural points where the story or topic shifts, creating chunks that feel more natural and coherent.
- **Scientific and technical documents:** These documents typically include specialized terms, equations, figures, tables, and carefully structured arguments or experimental steps. Effective chunking should respect these elements—for instance, keeping figures together with their captions or making sure detailed explanations aren't split randomly. Hybrid methods like S2-Chunking [16], which take visual layout into account, can be especially helpful for these types of documents.
- **Educational materials:** For instructional content, chunking often works best when aligned with learning goals or key concepts. Researchers have looked into using semantic similarity methods to divide educational texts into clear, meaningful modules, where each chunk matches a specific learning objective [23].

XII. Evaluation Strategies and Persistent Challenges

Evaluating how well document chunking methods work can be quite complicated. The main difficulty is that the "quality" of a chunk usually isn't something you can measure directly. Instead, it's best judged by how useful the chunks are for later NLP tasks. Because of this indirect way of measuring, and because there are so many different chunking methods and use cases, evaluations can vary widely and sometimes produce inconsistent results.

Downstream task performance: The most common way to evaluate chunking methods is by checking how they affect the performance of specific NLP tasks that follow chunking [18]. For example, in RAG systems, researchers usually measure the quality of the generated answers using metrics like ROUGE, BLEU, METEOR, or human ratings of accuracy and relevance. They might also evaluate how well the retrieval step works, using metrics such as precision, recall, Mean Reciprocal Rank (MRR), or Normalized Discounted Cumulative Gain (nDCG) [31] [32] [18] [24]. For summarization tasks, metrics like ROUGE are often used to compare summaries created from chunks against reference summaries [5] [33]. In Machine Reading Comprehension tasks, common evaluation metrics include F1 scores or Exact Match accuracy on question-answering datasets [15].

- **Critique/limitations:** A major limitation is that it's hard to separate the effect of chunking from other parts of the system (such as the embedding model used for retrieval, the language model used for generation, or

the specific dataset being used). Poor results might be caused by ineffective chunking, but they could also come from problems elsewhere in the pipeline [24]. To fairly compare different chunking methods, all other parts of the system need to stay exactly the same, which can be difficult to achieve across different studies. Additionally, good results on one task (like RAG question-answering) might not carry over to another task (like summarization or document classification), since different tasks might require different chunking approaches.

Intrinsic evaluation metrics: Some researchers have tried using intrinsic metrics, which directly measure the quality of chunks themselves, without relying on a specific downstream task. These metrics often look at things like semantic coherence within chunks (for example, the average similarity between sentences in the same chunk or topic coherence scores) or distinctiveness between chunks (such as low similarity between neighboring chunks). The main idea is that good chunks should be internally consistent in meaning and clearly different from each other.

- **Critique/limitations:** The main difficulty is that there's no clear agreement on what exactly makes a chunk "good" on its own or how to measure this reliably. For instance, while having high coherence within a chunk might seem ideal, overly coherent chunks could end up being too small and might not provide enough context for certain tasks. Additionally, it's often unclear how well these intrinsic metrics actually predict real-world performance on downstream tasks. The relationship between intrinsic measures and practical outcomes can be weak, unclear, or vary depending on the specific task, limiting their usefulness as standalone evaluation methods.

Novel evaluation frameworks: Realizing the shortcomings of current evaluation methods, researchers have started developing more detailed evaluation approaches. For example, Günther et al. [18] introduced a token-level relevance evaluation for retrieval in RAG systems. Instead of simply checking whether a retrieved chunk contains any relevant information (a yes-or-no measure), they measure what proportion of tokens within the retrieved chunks are actually relevant. This gives a more detailed picture of how well different chunking methods match retrieved content to the user's actual information needs. Their approach showed that different chunking strategies can lead to noticeable differences in token-level recall and precision.

- **Critique/limitations:** These newer evaluation methods often need specialized annotations (for example, marking exactly which tokens in a document are relevant to a specific query). Creating these annotations can be costly and time-consuming. As a result, these methods aren't yet widely used or standardized, making it difficult to compare results across different studies.

Broader Challenges in Evaluation: Aside from specific evaluation methods, there are several general challenges that remain when evaluating document chunking strategies:

1) **Lack of standardized benchmarks:** Currently, there aren't many widely accepted benchmark datasets or standard evaluation methods specifically created for comparing chunking approaches across different fields, document types, and tasks [18] [22]. This makes it hard to clearly determine which methods work best in general.
2) **Difficulty in comparison:** Because chunking methods vary greatly—from simple fixed-size splits to advanced LLM-based techniques—and involve many different parameters (like chunk length, overlap size, similarity thresholds, or embedding models), it's very challenging to fairly and thoroughly compare them [24].
3) **Cost- benefit trade- off:** As pointed out by Qu et al. [24] and others, it's important to weigh the performance improvements of advanced semantic chunking methods against their higher computational costs and complexity, especially when considering real-world use. However, this trade-off is often overlooked or not clearly reported in research studies.

XIII. Chunking within the Broader NLP Ecosystem: Interdependencies and Synergies

Document chunking is an important pre- processing step, but it doesn't work in isolation. How effective it is—and which chunking method works best—depends heavily on the needs and characteristics of other components in the NLP pipeline. This is especially clear in complex systems like Retrieval-Augmented Generation (RAG), document summarization, and Machine Reading Comprehension (MRC). Additionally, recent research is looking into new ways to refine or use chunked information beyond just the initial splitting and retrieval steps. This shows that our understanding of chunking's role is continuing to evolve.

A. Interaction with Downstream Tasks

The chunking method chosen directly affects the input given to downstream models, significantly impacting their performance. In RAG systems, as discussed earlier, chunking determines how detailed, coherent, and complete the retrieved knowledge pieces are, which the language model then uses to generate responses [18] [25]. Poorly created chunks can cause the retrieval of irrelevant pieces, incomplete details, or broken context, making it harder for the model to produce accurate, complete, and reliable answers [35]. Many evaluation studies looking at RAG performance [31] [32] [18] [24] have consistently highlighted this strong connection between chunking quality and the quality of the final output.

For long document summarization, chunking is often used as a practical way to handle texts that are too long for summarization models to process at once [5] [33]. Methods like Semantic Self-Segmentation (S3) [5] aim to create meaningful chunks that can each be summarized separately, with these smaller summaries later combined into a final summary. The quality of the initial chunks directly affects how clear, complete, and concise the final summary will be. Poor

chunking can split or lose important information, resulting in summaries that feel incomplete or disconnected.

In Machine Reading Comprehension (MRC) tasks involving long texts, chunking determines how the document is presented to the model, which needs to find and extract answers from the given context. Simple fixed-size chunking can accidentally split answers or relevant context across different chunks, making it harder for the model to find the complete answer. This issue has led to more advanced methods like Recurrent Chunking Mechanisms [15], which allow context to flow between chunks, helping the model better understand the entire document as it processes each chunk in sequence.

B. Beyond Initial Segmentation and Retrieval: Post-Retrieval Refinements:

Processing chunked information doesn't always end after the initial retrieval step. Since even advanced chunking and retrieval methods can sometimes return noisy, repetitive, or partly irrelevant chunks, researchers have suggested additional refinement steps after retrieval. For example, ChunkRAG [23] adds an LLM-based filtering stage after retrieval but before generation. This filter checks the relevance and quality of retrieved chunks, removing those that aren't useful, so the generator model receives cleaner and more focused information. The goal is to improve the accuracy and factual correctness of the final output by reducing noise.

Other studies focus on creating shorter, more informative representations from retrieved chunks. For instance, CHULO (Chunk-Level Key Information Representation) [12] uses an LLM to summarize key points from each chunk, creating condensed versions. This helps downstream models handle long contexts more efficiently, reducing computational load and allowing the model to focus on the most important information.

These developments show a changing view of chunking—not just as an initial segmentation step, but as the start of a multi-step information processing pipeline. Improving overall system performance might involve not only better initial chunking methods, but also additional stages like filtering, reranking, summarizing, or transforming chunks before they're used by the final model. This reflects a growing recognition that simply retrieving relevant chunks may not be enough for complex reasoning and generation tasks, and that further refinement can significantly improve results.

Critiques/ limitations: Although post-retrieval refinement methods show promise, they add extra complexity and computational costs to the NLP pipeline. For example, ChunkRAG [13] requires an additional LLM inference step to filter retrieved chunks, increasing latency and operational expenses. Similarly, CHULO [12]depends on LLM inference to create condensed summaries. The success of these refinement steps also heavily depends on the quality of the initial chunks and the performance of the models used for filtering or summarizing. Additionally, there's a risk of losing important information if the filtering or summarization is too aggressive or inaccurate.

XIV. Discussion

This literature review has explored the evolving field of document chunking, an important pre- processing step that helps Large Language Models (LLMs) effectively handle long texts [3] [5] [9] [12] [17]. Good chunking strategies are especially important because current LLMs have limited context lengths, and the quality of chunks directly affects the accuracy and relevance of outputs in systems like Retrieval-Augmented Generation (RAG) [11] [31] [30] [18] [20] [25]. Our review shows a clear progression from simple, rule-based methods to more advanced, meaning-based and adaptive approaches, each trying to balance semantic quality, context preservation, and computational efficiency.

We identified several main categories of chunking methods (RQ1). Basic methods, like fixed-size chunking [3] [12] [14] [18] [29] [24] or simple rule-based splits (by sentence or paragraph) [7] [30] [22] [29], are easy to implement and fast, but often break context and meaning, limiting their usefulness for complex documents or tasks. While these methods are useful as baselines, they often fall short when semantic accuracy is important.

To address these limitations, semantic chunking methods have become popular (RQ2). These methods use embeddings, clustering, or dynamic programming to create chunks based on meaning [3] [5] [7] [12] [23] [34] [33]. The goal is to produce chunks that are internally coherent and contextually meaningful, improving downstream tasks like summarization or question-answering. However, our review found that while these methods seem promising, the actual performance gains over simpler methods aren't always clear or consistent, especially considering their higher computational costs and the complexity of tuning parameters [3] [24]. Thus, deciding whether to use semantic chunking depends heavily on the specific context and available resources.

More advanced and dynamic chunking methods (RQ2) represent the cutting edge of current research. Techniques like Recurrent Chunking Mechanisms [15] and "Late Chunking" [20] directly address the problem of losing context across chunk boundaries. Methods that use LLMs directly for chunking, such as LumberChunker [31], perplexity-based chunking [32], or dynamic granularity approaches like Mix-of-Granularity (MoG) [21], offer greater flexibility and adaptability. Hybrid methods, combining structural and semantic information (like S2-Chunking [16] or specialized financial report chunking [11]), also show promise, especially for structured documents. However, these advanced methods often come with increased complexity, higher computational costs, and reliance on powerful models.

Our review emphasizes that the best chunking method depends strongly on the type of document and the specific NLP task (RQ3, RQ4). Legal texts [22], financial reports [11] [20], narrative stories [31], and educational materials [23] each have unique characteristics that require tailored chunking strategies. This suggests that no single chunking method will work best everywhere. Instead, choosing and tuning a chunking strategy

TABLE I
COMPARISON OF CHUNKING STRATEGIES

Category	Approach Examples	Key Characteristics	Pros	Cons	Key References
Foundational	Fixed-Size, Rule-Based (Sentence/Paragraph)	Simple heuristics, structural delimiters (e.g., punctuation, section breaks)	Computationally cheap, easy to implement	Semantically naive, context fragmentation, inflexible, performance highly parameter-dependent on parameters	[3], [7], [12], [18], [22], [24], [29], [30], [35]
Semantic (Embedding-based)	Similarity Thresholds, Clustering, S3	Uses word/sentence embeddings to identify semantic shifts or group related content	Aims for better semantic coherence and contextually relevant chunks, potentially improving downstream task performance	Computationally more intensive than foundational methods, sensitive to embedding quality and parameter tuning (e.g., thresholds, number of clusters); benefits over well- tuned traditional methods not always consistently demonstrated, risk of over/under-segmentation	[3], [5], [7], [12], [17], [23], [24], [33], [34]
Advanced & Dynamic	RCM, Late Chunking, LLM-driven (LumberChunker, PPL), MoG	Leverages LLMs, dynamic granularity based on context or query, explicit context preservation mechanisms across chunks	High adaptability to content nuances and query-specific tasks; potential for superior content handling and more meaningful segmentation	Very high computational cost and latency, especially for LLM- driven methods; dependency on powerful models and complex prompt engineering; can be harder to interpret and debug	[15], [18], [20], [21], [31], [32]
Hybrid	S2-Chunking, Financial Report Chunking	Combines semantic and structural/layout/visual features of the document	Effective for structured or semi-structured documents (e.g., PDFs, financial reports, scientific papers) by leveraging multiple information sources for more robust segmentation	Increases complexity in design and implementation; often domain-specific, requiring adaption or retraining for new document types or layouts; feature engineering can be intensive	[11], [16], [20], [29]
Post-Retrieval Refinement	ChunkRAG, CHULO	Processes chunks *after* retrieval but *before* final use (e.g., filtering irrelevant chunks, condensing information with LLMs.)	Improves precession and efficiency of downstream tasks by providing cleaner, more focused input to the generator; mitigates some issues of imperfect initial chunking	Adds further computational steps, letency, and complexity to the pipeline;risk of information loss during refinement if not carefully tuned; effectiveness depends on he quality of the refinement model	[12], [13]

requires careful consideration of the data and the task at hand.

A major ongoing challenge is evaluating chunking methods effectively (RQ5). Currently, there aren't widely accepted benchmarks or intrinsic metrics that reliably predict downstream performance. Most evaluations rely on measuring the impact on end-to-end tasks (like RAG question-answering or summarization), making it hard to isolate the specific contribution of chunking [24]. Newer evaluation methods, such as token-level relevance [18], are promising but not yet widely adopted. Developing standardized evaluation frameworks and diverse benchmark datasets is crucial for future progress. Additionally, researchers need to pay more attention to the trade-off between performance improvements and computational costs [24].

This review has important implications for both researchers and practitioners. Researchers should continue developing chunking methods that balance semantic quality, context preservation, and computational efficiency. Creating standardized evaluation methods and benchmarks is essential for meaningful comparisons and systematic progress. Practitioners should carefully choose chunking strategies based on their specific data, tasks, and available resources, rather than relying on a "one-size-fits-all" approach. Additionally, recent research into post-retrieval refinement methods [12] [13] suggests that chunking should be viewed as part of a broader information-processing pipeline, where initial segmentation can be improved by additional filtering, summarizing, or re-ranking steps.

In short, while chunking methods have evolved significantly—from simple splits to intelligent, adaptive segmentation—the search for optimal approaches continues. Balancing semantic accuracy, context completeness, computational effi-

ciency, and task-specific needs remains a key challenge and an active area of research.

XV. Limitations

While this literature review aimed to provide a thorough overview of document chunking methods, it has several limitations that should be kept in mind when interpreting the results.

First, although we used a structured search across major databases (Web of Science, arXiv, Scopus, ACL Anthology) and Google Scholar, our literature retrieval wasn't exhaustive. Given how quickly NLP research evolves—especially around Large Language Models and chunking—some very recent studies (after December 2024) or papers published in specialized venues might have been missed. Additionally, our keyword-based search might have unintentionally overlooked relevant studies that used different terminology or introduced new concepts not yet widely recognized.

Second, our inclusion criteria limited the review to English-language publications. This means valuable research from non-English-speaking academic communities might have been excluded, potentially affecting the global representativeness of our findings. Also, while including preprints (like those from arXiv) allowed us to cover cutting-edge research, some of these studies may not have undergone full peer review, which could affect the reliability of their claims.

Third, the process of categorizing and synthesizing the literature involved some subjective interpretation. Although we tried to be consistent and objective, different researchers might categorize methods or interpret findings differently.

Fourth, our review focused mainly on semantic and advanced chunking methods due to their growing importance for LLMs. While we discussed simpler methods to provide context, we didn't explore every detail or variation of basic chunking approaches. As a result, some specific applications or optimizations of simpler methods might have received less attention.

Fifth, evaluating chunking methods is inherently challenging, as highlighted in our review. Many studies measure chunking effectiveness indirectly through downstream tasks, making it difficult to isolate the exact impact of chunking from other system components (like embedding models or LLM architectures). The lack of standardized benchmarks and widely accepted intrinsic metrics also makes direct comparisons between studies difficult.

Finally, the rapid pace of innovation in NLP means the field of document chunking is constantly changing. This review provides a snapshot based on literature available up to December 2024. New methods, evaluation approaches, and insights will likely emerge quickly, potentially shifting our understanding of the methods discussed here.

These limitations don't diminish the value of this review but provide transparency about its boundaries. Future research could address these limitations by regularly updating the review, expanding linguistic coverage, or contributing to standardized evaluation frameworks for document chunking.

XVI. Conclusion

This literature review has carefully examined different document chunking methods, highlighting important advances and various approaches aimed at improving information retrieval and enhancing Retrieval-Augmented Generation (RAG) systems. Throughout the review, we explored several key techniques, including fixed-size chunking, semantic chunking, dynamic and hierarchical methods, and hybrid approaches.

Fixed-size chunking is simple and easy to implement, but it often struggles to preserve context effectively. Semantic chunking methods, on the other hand, produce more meaningful and context-rich chunks, though they typically require more computational resources. Hybrid methods combine the strengths of both approaches, balancing performance and efficiency.

Newer methods, such as dynamic and multimodal chunking, show great promise by adapting chunk boundaries based on content, user queries, or context. These innovative approaches can significantly improve the relevance and coherence of retrieved information.

However, our review also identified important limitations. Semantic and multimodal chunking methods often come with high computational costs, and there is currently a lack of standardized benchmarks to fairly evaluate and compare chunking performance across different tasks and document types.

Future research should focus on creating more efficient chunking algorithms, developing standardized evaluation metrics, and designing specialized chunking strategies for specific fields like medicine, law, or technical documentation. These targeted improvements will help make chunking methods more practical and effective for real-world applications.

In summary, while significant progress has been made, continued research and refinement are essential to fully realize the potential of document chunking methods. Such efforts will ultimately lead to more accurate and contextually relevant information retrieval in a wide range of knowledge-intensive applications.

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[2] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, *et al.*, "Language models are few-shot learners," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 1877–1901, 2020.

[3] E. Muszyńska, *Semantic chunking*. PhD thesis, University of Cambridge, 2020. https://www.repository.cam.ac.uk/items/8bd11d65-bbb3-4b1b-b84e-7d41164f415c.

[4] Y. Li, S. C. Han, Y. Dai, and F. Cao, "CHULO: Chunk-level key information representation for long document processing," 2024.

[5] G. Moro and L. Ragazzi, "Semantic self-segmentation for abstractive summarization of long documents in low-resource regimes," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, pp. 11990–11998, Jun 2022.

[6] I. Beltagy, M. E. Peters, and A. Cohan, "Longformer: The long-document transformer," 2020.

[7] S. Todeschini, "How to chunk text data - a comparative analysis," Jul 2023. https://towardsdatascience.com/how-to-chunk-text-data-a-comparative-analysis-3858c4a0997a.

[8] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, A. Fan, V. Chaudhary, T. Rocktäschel, D. Kiela, *et al.*, "Retrieval-augmented generation for knowledge-intensive nlp tasks," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.

[9] Superteams.ai, "A deep-dive into chunking strategy, chunking methods, and precision in rag applications," Mar 2025. https://www.superteams.ai/blog/a-deep-dive-into-chunking-strategy-chunking-methods-and-precision-in-rag-applications.

[10] J. Bohannon, "Chatgpt cited nonexistent legal cases. a judge is not amused," May 2023. https://www.science.org/content/article/chatgpt-cited-nonexistent-legal-cases-judge-not-amused.

[11] A. Jimeno Yepes, Y. You, J. Milczek, S. Laverde, and L. Li, "Financial report chunking for effective retrieval augmented generation," 2024.

[12] Nirdiamant, "Semantic chunking: Improving ai information retrieval," Sep 2024. https://medium.com/@nirdiamant21/semantic-chunking-improving-ai-information-retrieval-2f468be2d707.

[13] I. S. Singh *et al.*, "Chunkrag: Novel llm-chunk filtering method for rag systems," 2024.

[14] K. Kutumbe, "Mastering chunking for better semantic retrieval with large language models," Dec 2024. https://kshitijkutumbe.medium.com/mastering-chunking-for-better-semantic-retrieval-with-large-language-models-6a6f1fa5f220.

[15] H. Gong, Y. Shen, D. Yu, J. Chen, and D. Yu, "Recurrent chunking mechanisms for long-text machine reading comprehension," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 6751–6761, 2020.

[16] P. Verma, "S2 chunking: A hybrid framework for document segmentation through integrated spatial and semantic analysis," 2025.

[17] Superlinked, "Semantic chunking," Oct 2024. https://superlinked.com/vectorhub/articles/semantic-chunking.

[18] B. Smith and A. Troynikov, "Evaluating chunking strategies for retrieval," tech. rep., Chroma, Jul 2024. https://research.trychroma.com/evaluating-chunking.

[19] A. V. Duarte, J. Marques, M. Graça, M. Freire, L. Li, and A. Oliveira, "Lumber chunker: Long-form narrative document segmentation," 2024.

[20] M. Günther, I. Mohr, D. J. Williams, B. Wang, and H. Xiao, "Late chunking: Contextual chunk embeddings using long-context embedding models," 2024.

[21] Z. Zhong, H. Liu, X. Cui, X. Zhang, and Z. Qin, "Mix-of-granularity: Optimize the chunking granularity for retrieval-augmented generation," 2025.

[22] A. F. Ferraris, D. Audrito, G. Siragusa, and A. Piovano, "Legal chunking: Evaluating methods for effective legal text retrieval," in *Legal Knowledge and Information Systems (JURIX 2024)*, IOS Press, 2024.

[23] K.-N. Tran *et al.*, "Document chunking and learning objective generation for instruction design," 2018.

[24] R. Qu, F. Bao, and R. Tu, "Is semantic chunking worth the computational cost?," *Qeios*, Nov 2024. Preprint available at arXiv:2410.13070.

[25] C. Grover and R. Tobin, "Rule-based chunking and reusability," in *Proceedings of LREC'06*, 2006.

[26] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, *et al.*, "The prisma 2020 statement: An updated guideline for reporting systematic reviews," *BMJ*, vol. 372, 2021.

[27] N. C. Vasireddy, "Supplementary material for 1 for this paper." https://drive.google.com/file/d/1EnKtazW2n8RUIW-cIrzshpn8OEDiEly0/view, 2025. Accessed: 2025-05-06].

[28] N. C. Vasireddy, "Supplementary material 2 for "semantic chunking strategies in rag systems"." https://drive.google.com/file/d/1FCjvImfLuHZlIoeocseyJ_X-fSokKars/view?usp=drivesdk, 2025. Accessed: 2025-05-06.

[29] S. Klink and T. Kieninger, "Rule-based document structure understanding with a fuzzy combination of layout and textual features," *International Journal on Document Analysis and Recognition (IJDAR)*, vol. 3, pp. 188–200, Nov 2000.

[30] I. Böhm, "Rule based vs. statistical chunking of conll data sets." Unpublished manuscript, Jan 2005.

[31] A. V. Duarte, J. Marques, M. Graça, M. Freire, L. Li, and A. Oliveira, "Lumber chunker: Long-form narrative document segmentation," Jun 2024.

[32] J. Zhao, Z. Ji, Y. Feng, P. Qi, S. Niu, B. Tang, F. Xiong, and Z. Li, "Meta-chunking: Learning efficient text segmentation via logical perception," Nov 2024.

[33] A. Gidiotis and G. Tsoumakas, "A divide-and-conquer approach to the summarization of long documents," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 1–13, Sep 2020.

[34] K. Hacioglu, "A lightweight semantic chunking model based on tagging," in *Proceedings of the Human Language Technology Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations*, pp. 209–212, 2004.

[35] G. Kuratomi, P. Pirozelli, F. G. Cozman, and S. M. Peres, "A rag-based institutional assistant," Jan 2025.